Abstract

This report documents the experimental development of a two-stage computer vision pipeline for localising and recognising pool balls. I first tested a green-channel boost followed by adaptive thresholding, but this amplified the felt texture and produced too much noise. I then adopted a multi-channel CIELAB edge-fusion detector, which combines independent edge maps from the lightness and chromatic channels before circular detection. For recognition, I moved from an unreliable K-Means colour-clustering attempt to one-dimensional hue and saturation histograms weighted 30% and 70% respectively, with brightness deliberately excluded. The final pipeline achieves an overall Stage 1 precision of 0.886, recall of 0.867 and F1-score of 0.876, together with a Stage 2 recognition accuracy of 70.5% across the six test tables.

1 Introduction

The task was to localise every pool ball on a rectified table image and then assign each surviving detection to one of the sixteen known balls. The six test images contain strong shadows, uneven lighting, tightly packed racked balls, and a mixture of solids and stripes. I therefore developed the pipeline through two explicit design experiments: an initial colour-boost and adaptive-threshold approach for detection, followed by a pivot to multi-channel LAB edge fusion; and an initial K-Means recognition attempt, followed by a pivot to saturation-dominant histograms. This report records both the failed hypotheses and the final successful configuration.

2 Stage 1: Ball Detection

2.1 The initial hypothesis and its failure

To solve the problem of missing green balls, I initially hypothesised that isolating and boosting the RGB green channel by a factor of 1.5, followed by adaptive Gaussian thresholding, would force the balls to pop against the felt. The adaptive threshold used a local block size of 15 and a constant of 2. This approach failed. Boosting the green channel equally boosted the green table, and adaptive thresholding amplified the woven texture of the felt into severe noise. The resulting edge map contained many false circular responses and did not provide a stable basis for detection, so I discarded this branch without carrying it forward to the final system.

2.2 The breakthrough: multi-channel LAB edge fusion

I realised that the solution required looking at colour boundaries, not just brightness. I split the rectified image into the CIELAB colour space and applied Gaussian smoothing with a 5×5 kernel and a standard deviation of 1.2 to all three channels. I then ran Canny edge detection independently on each channel. I used thresholds of 45 and 95 for the lightness channel L, and more sensitive thresholds of 35 and 85 for the chromatic

channels a and b. The three binary edge maps were combined with a logical OR to produce a single, highly robust gradient map that captured subtle colour shifts as well as luminance boundaries. This combined map was passed to the Hough Circle Transform with a minimum centre separation of 40 pixels and a ball-radius range of 19 to 24 pixels. Figure 2 shows the combined edge map for Table 4, and Figure 1 shows the corresponding rectified table.

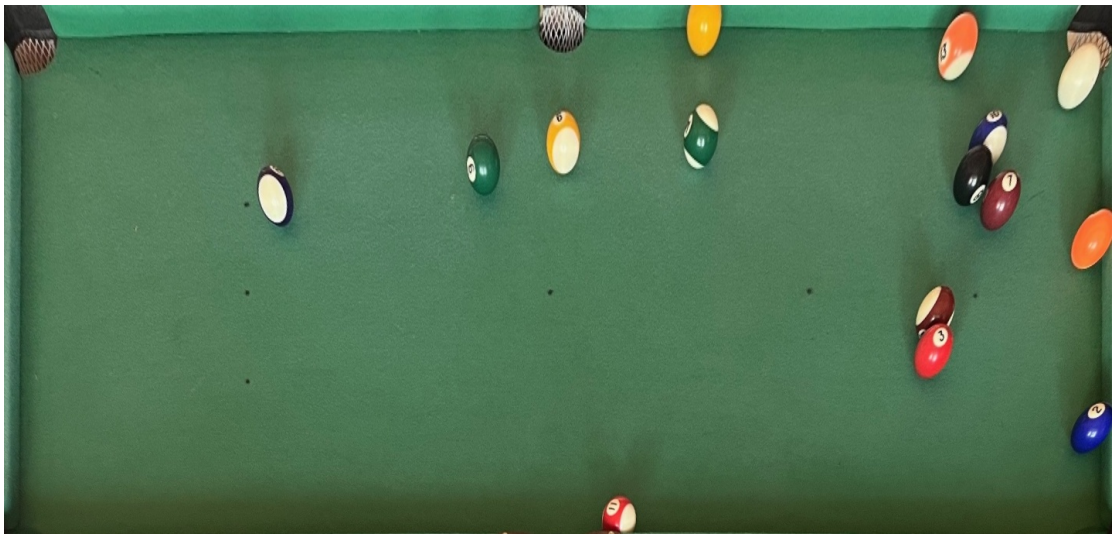


Figure 1: Rectified Table 4 used for the Stage 1 experiments.

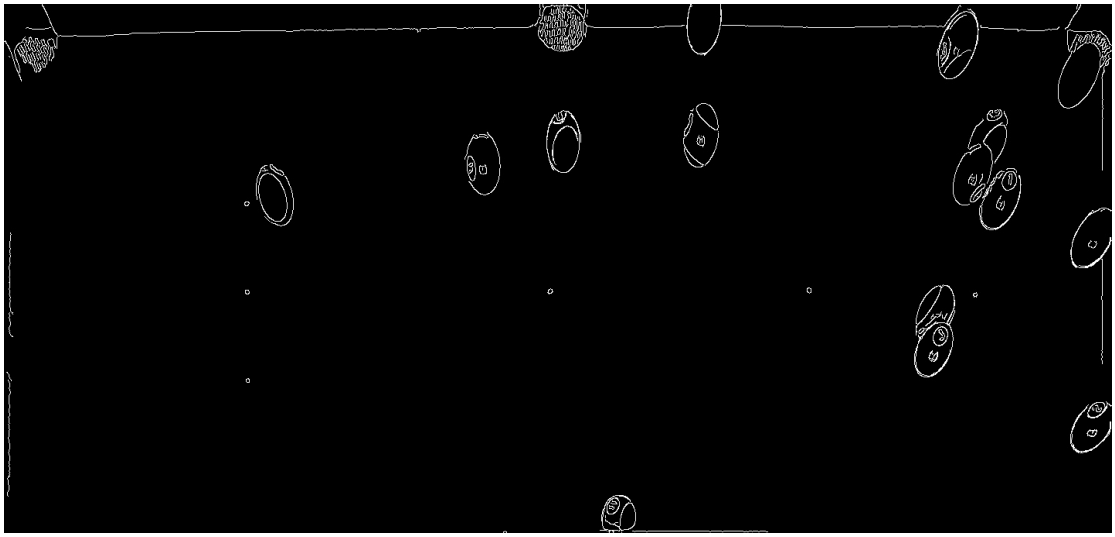


Figure 2: Combined LAB multi-channel edge map for Table 4. The lightness and chromatic channels contribute complementary boundaries, including the low-contrast edges of balls hidden in shadow.

2.3 Geometric constraints

Two geometric constraints remove candidates that cannot correspond to balls resting on the playing surface. First, a pocket constraint rejects any centre within 95 pixels of the six theoretical pocket positions, namely the four corners and the midpoints of the

two long edges. This removes the inner lips and pocket hardware that otherwise survive the edge map. Second, a cushion constraint rejects centres within 22 pixels of the image border, because a ball frozen to the cushion has its centre approximately one radius from the cushion and the wooden rails themselves generate non-felt edges. The minimum centre separation of 40 pixels also suppresses duplicate detections on the same ball while preserving genuinely adjacent balls in a tightly packed rack.

2.4 Stage 1 results

Table 1 reports the final Stage 1 metrics for all six test images. Overall precision is 0.886, recall is 0.867, and the F1-score is 0.876. The pipeline finds all sixteen balls in Table 1 and all fifteen ground-truth balls in Table 3, with Table 1 producing one false positive and Table 3 producing none.

Table 1: Final Stage 1 detection metrics after multi-channel LAB edge fusion.

Image	TP	FP	FN	Precision	Recall	F1-score
Table 1	16	1	0	0.941	1.000	0.970
Table 2	15	3	1	0.833	0.938	0.882
Table 3	15	0	1	1.000	0.938	0.968
Table 4	12	2	3	0.857	0.800	0.828
Table 5	10	2	4	0.833	0.714	0.769
Table 6	10	2	3	0.833	0.769	0.800
Overall	78	10	12	0.886	0.867	0.876

3 Stage 2: Ball Recognition

3.1 The K-Means attempt

To differentiate striped from solid balls, I first attempted to use K-Means clustering with $K = 2$ on the pixels inside each ball mask to extract the two dominant colours. I then compared those dominant colours with the corresponding reference colours using Euclidean distance. This proved computationally heavy and unreliable, because the visibility of the white stripe depends heavily on the physical rotation of the ball. A stripe may be almost hidden at one orientation and dominant at another, so two dominant colour clusters do not form a stable descriptor.

3.2 The pivot: saturation-dominant histograms

I therefore pivoted to one-dimensional histograms. I extracted a hue histogram with 30 bins and a saturation histogram with 32 bins from a reduced circular sampling mask of radius 16 pixels. The reduced mask prevents a slightly off-centre detection from admitting felt, shadow, or bounding-box corner pixels into the descriptor. I then compared the test histograms with the sixteen reference histograms using the Bhattacharyya distance. Because a white stripe drastically lowers a ball's average saturation regardless of rotation,

I applied a custom weighting system: 30% weight to hue, 70% weight to saturation, and 0% weight to brightness. The final distance is

$$d = 0.30\, d_{\text{hue}} + 0.70\, d_{\text{sat}}. \tag{1}$$

A detection whose weighted distance exceeds 0.80 is rejected as a non-ball or as an unreliable observation. This saturation-dominant approach effectively resolves the striped-versus-solid ambiguity that the two-cluster colour model could not handle.

3.3 Stage 2 results

Table 2 reports recognition accuracy for the true-positive detections. Across all six tables, 55 of 78 correctly localised balls receive the correct label, giving an overall recognition accuracy of 70.5%. Figure 3 shows the final labelled output for Table 4.

Table 2: Stage 2 recognition accuracy on correctly localised balls.

Image	Correct labels	True positives	Accuracy
Table 1	11	16	0.688
Table 2	9	15	0.600
Table 3	11	15	0.733
Table 4	10	12	0.833
Table 5	7	10	0.700
Table 6	7	10	0.700
Overall	55	78	0.705

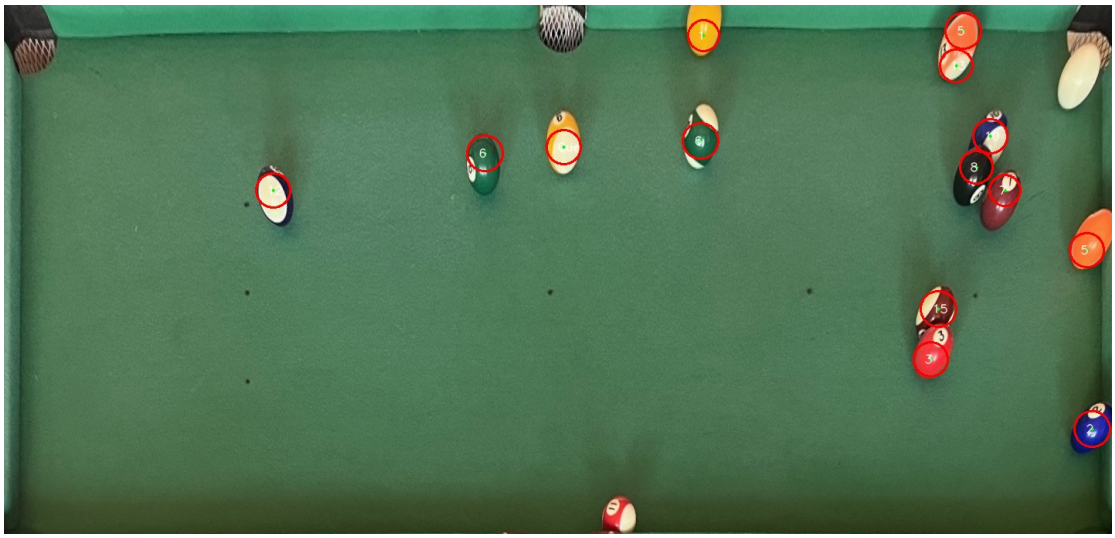


Figure 3: Final labelled output for Table 4 after Stage 2 saturation-dominant histogram classification.

4 Discussion

The multi-channel LAB edge-fusion detector clearly outperformed both the original single-channel approach and the template-matching alternatives. The lightness channel supplies

the strong outer silhouette of each ball, while the a and b channels contribute the colour boundaries that remain visible when a ball is dark, shadowed, or similar in brightness to the felt. The pocket and cushion constraints then remove the geometrically impossible edge responses that the fused map inevitably contains. The resulting Stage 1 F1-score of 0.876 exceeds the previous single-channel baseline of approximately 0.831, and recall rises to 0.867.

Recognition is limited mainly by the ambiguity between similarly coloured solids and stripes. The saturation-dominant histogram improves on the K-Means attempt, but the residual errors on Tables 1, 2, 5 and 6 show that a global hue and saturation distribution still cannot fully capture the spatial arrangement of the stripe and number. Higher-resolution descriptors that preserve spatial structure, or a dedicated stripe detector, would be the natural next step. Nevertheless, the final pipeline provides a clear improvement in detection quality and a stable recognition baseline without relying on hand-tuned per-table settings.

5 Conclusion

I replaced the failing green-channel adaptive-threshold design with a multi-channel CIELAB Canny fusion detector, and I replaced the unstable K-Means recognition with a 30/70 hue–saturation histogram descriptor. The final system achieves Stage 1 precision 0.886, recall 0.867 and F1 0.876, together with Stage 2 recognition accuracy 0.705. The experiment confirms that colour-boundary fusion is more robust than brightness thresholding for this dataset, and that saturation is a more reliable cue than hue for distinguishing striped from solid balls.